

Wall–Tap Delay after the Complete Assembly

A Physical Lock between Observation Law, Operational Duration, Medium-Anchored Distance, and the Theorem Ladder of Resolution Geometry

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Abstract

The wall–tap observation was the historical seed of Resolution Geometry (RG): an emitted event was followed by a later confirmation, and changes of position, wall and rhythm changed what remained distinguishable. The original reconstruction correctly replaced retrospective precision by a declared acoustic experiment, but it preceded the complete theorem ladder, operational duration, physical-distance identification and the kernel–stabilizer assembly. This paper gives the mature physical version. The primitive datum is an ordered pair of clock events, not a distance. Raw latency is a sum of trigger, source, path, transducer and detector terms. After nuisance closure, the propagation duration is a natural module on the joint law. A single acoustic return identifies a slowness-weighted path, not metres; in a homogeneous medium it fixes only d/c_{air} . Metres enter only after a declared and independently controlled medium-speed anchor. Moreover, the familiar factor $1/2$ is valid as a wall distance only for the collocated monostatic normal-return geometry; a general source–wall–microphone record is a bistatic path-sum constraint. Multiple labelled returns, closure relations, regularity, path identity, environmental telemetry and held-out anchors then lock a local physical geometry. The rhythm transition is not a universal 30 ms boundary: it scales with the measured response tail and the declared decision statistic. A deterministic hostile battery (seed 20260929) passes 12/12 registered checks and survives seeds 1, 42, 777 and 2026. It establishes three no-go results and a conditional positive theorem: wall–tap delay becomes physical distance exactly on certified windows where latency decomposition, medium scale, path model, label closure, regularity, stationarity and held-out validation lock jointly. The paper inserts this route into the complete RG theorem ladder without claiming historical instrumental precision, a universal auditory threshold, room reconstruction from one echo, or a new propagation law.

Contents

1	Question and change of status	2
2	Joint law of the mature experiment	2
3	Duration comes before distance	3
4	What one echo determines	3
5	Medium anchor and the acoustic metre	3
6	Multiple returns, closure and path identity	4
7	Resolution, rhythm and the childhood transition	5
8	Uncertainty floor and held-out validation	5

9	The Wall–Tap Physical Lock Theorem	6
10	Attack ledger	6
11	Non-claims	7
12	Conclusion	7

1 Question and change of status

The earlier wall–tap paper [1] established a complete protocol around trigger-relative acoustic returns, observable quotients, covariance, projectable transport and an entropy quotient. It deliberately stopped short of identifying delay with physical distance. The later ladder and physical-distance analysis [2, 4, 5] now make the missing route precise.

The mature question is not “was a delay perceived?” but:

Under which declared observation system does an ordered trigger–return interval descend to operational duration, when does that duration identify an acoustic path, and under which additional anchors and locks does that path become a distance in metres?

The answer has four typed stages:

event order $\longrightarrow$ calibrated duration $\longrightarrow$ slowness-weighted path $\longrightarrow$ physical distance.

None of the arrows is automatic.

2 Joint law of the mature experiment

Definition 2.1 (Wall–tap physical slice). A wall–tap slice is a joint observation system

$$\mathcal{S}_{\text{WT}} = (X, F_{\text{tot}}, \Psi), \quad F_{\text{tot}} = (F_{\text{trig}}, F_{\text{clk}}, F_{\text{wav}}, F_{\text{env}}, F_{\text{geom}}, F_{\text{ctrl}}),$$

where the channels record the trigger, calibrated clock, raw waveform, environmental state, declared source–receiver geometry and hostile controls. The kernel groupoid is

$$\mathcal{K}_{\text{WT}} = X \times_{F_{\text{tot}}} X \rightrightarrows X, \quad Q_{\text{WT}} = X/\mathcal{K}_{\text{WT}}.$$

A minimal latent state contains room geometry, boundary response, medium state, source waveform and onset, receiver response, trigger offset, clock map, reflection assignment and model discrepancy. Human memory or localization reports are separate protocols and are never inferred from the acoustic channel.

Proposition 2.2 (Natural descent of the acoustic modules). *On every regular stratum, trigger-relative duration, labelled path delays, normalized response features, closure residuals and their covariance are natural modules of the joint law. They therefore descend to Q_{WT} and are invariant under its effective symmetry group. A hidden device or path-assignment variable that changes one of these outputs is not gauge and must remain outside the observational kernel.*

Proof. Each listed object is computed from the joint law and is invariant under the declared relabellings. The natural-descent theorem of the kernel–stabilizer assembly [5] applies. A transformation changing an output does not lie in the kernel by definition. $\square$

3 Duration comes before distance

For trial i , the measured latency decomposes as

$$\Delta t_i^{\text{raw}} = \tau_{\text{trig},i} + \tau_{\text{src},i} + \tau_{\text{path},i} + \tau_{\text{trans},i} + \tau_{\text{det},i} + \varepsilon_i. \quad (1)$$

Theorem 3.1 (Wall–tap duration lock). *Let T be a trigger-relative clock readout on a regular history stratum. If calibration closes the nuisance kernel in (1), repeated devices have equal clock kernels, and the calibrated clock flow lies in the stabilizer of the joint law, then the path-duration charts form one clock groupoid with forced cocycle. Their transitions are affine under the clock flow. The resulting τ_{path} is an operational duration in the sense of [3]; without nuisance closure, only Δt^{raw} is observed.*

No-Go 3.2 (Raw latency is not propagation duration). No theorem can identify Δt^{raw} with τ_{path} from the acoustic return alone. Adding a common detector latency changes the former while preserving the path. Battery check WT01 recovers the injected 14 ms path duration only after nuisance subtraction; the naive latency retains about 7 ms of apparatus bias.

This is the first ladder lock: the wall–tap observation enters the duration branch before it enters the distance branch.

4 What one echo determines

For a homogeneous medium with speed c_m and a monostatic round trip to a plane wall,

$$\tau_{\text{path}} = \frac{2d}{c_m}. \quad (2)$$

For a source s , receiver m and specular point w on a boundary,

$$\tau_{\text{path}} = \frac{\|s - w\| + \|w - m\|}{c_m}. \quad (3)$$

In an inhomogeneous medium the primitive is more generally integrated slowness,

$$\tau_{\text{path}} = \int_{\gamma} \frac{ds}{c(x, \omega, t)}. \quad (4)$$

No-Go 4.1 (Delay–speed fibre; WT02). A single value of τ does not identify d . In (2), every pair (d, c_m) with $d/c_m = \tau/2$ has the same law. For $\tau = 12$ ms, speeds 330, 343 and 360 m/s give compatible distances 1.98, 2.058 and 2.16 m. The unit fibre is exact until c_m is anchored.

No-Go 4.2 (The half-factor is typed; WT11). The map $d = c_m \tau/2$ returns a wall distance only for the monostatic normal-return model. For general bistatic geometry it returns half a broken path length. In the registered control, the half path is approximately 3.04 m while the source’s normal distance to the wall is 1 m.

Thus the physical-distance rung must name the path model. “Echo distance” without this type is incomplete.

5 Medium anchor and the acoustic metre

The SI radar branch uses the exact vacuum speed c . Acoustic propagation does not. The speed of sound is a medium property and must be obtained through a declared thermodynamic or direct calibration route [8, 7, 9].

Definition 5.1 (Medium lock). A window has medium lock if the declared environmental and dispersion channels make all parameters affecting (4) identifiable within a registered bound. In the homogeneous-air branch this means a certified value

$$c_{\text{air}} = c_{\text{air}}(T, H, p, \omega)$$

with propagated covariance and a held-out check.

Proposition 5.2 (Conditional conversion to metres; WT03 and WT06). *On a medium-locked monostatic window,*

$$\hat{d} = \frac{\hat{c}_{\text{air}} \hat{\tau}_{\text{path}}}{2}$$

is a physical-distance estimator. Its first-order relative uncertainty is

$$\frac{\delta d}{d} = \frac{\delta c_{\text{air}}}{c_{\text{air}}} + \frac{\delta \tau_{\text{path}}}{\tau_{\text{path}}} + \text{extcrossandmodelterms}.$$

The battery recovers the declared ranges with median relative error below 0.3%; a five-degree temperature miscalibration produces a visible common scale bias. WT06 verifies the first-order proportionality.

The metre is not extracted from Fisher geometry. It enters through a traceable medium-speed calibration and the measured clock duration. This is fully consistent with the physical-distance no-go and anchor theorem [4].

6 Multiple returns, closure and path identity

Consider a rectangular room of known extents W, H , a collocated source–receiver at (x, y) , and four labelled first-order returns. On the monostatic branch,

$$d_L = x, \quad d_R = W - x, \quad d_B = y, \quad d_T = H - y, \quad d_k = \frac{c_m \tau_k}{2}.$$

Hence

$$d_L + d_R = W, \quad d_B + d_T = H, \tag{5}$$

and

$$\hat{x} = \frac{d_L + W - d_R}{2}, \quad \hat{y} = \frac{d_B + H - d_T}{2}.$$

Theorem 6.1 (Local wall–tap geometry lock). *On a window satisfying:*

1. *duration lock;*
2. *medium lock;*
3. *a declared monostatic or bistatic path model;*
4. *labelled paths with closure residuals inside a registered tolerance;*
5. *constant target and nuisance rank;*
6. *held-out anchor validation;*

the labelled path map descends to a regular physical-distance chart. For the rectangular monostatic branch its Jacobian has rank two and (5) supplies two independent model checks. The metric scale is the medium-anchored metre, not the distinguishability scale.

WT04 obtained rank two, millimetre-scale synthetic reconstruction error and closure residuals below the registered bounds. The theorem does not extend across path births, merged peaks, label swaps or unresolved multipath: those are walls of the certified atlas.

No-Go 6.2 (Permutation fibre; WT05). Removing path labels does not merely increase variance. It leaves a finite exact fibre: several positions satisfy the same unordered multiset of returns and the room closure equations. Therefore path identity is an independent gate, not a cosmetic preprocessing choice.

7 Resolution, rhythm and the childhood transition

The childhood observation included a change when taps became too frequent for returns to remain isolated. Mature RG treats this as an escalation problem in the history law, not as a universal perceptual constant.

Let $h(t)$ be the measured impulse response and

$$y_T(t) = \left(\sum_n \delta(t - nT) \right) * h(t).$$

For a declared statistic $\chi(T)$ comparing isolated-return energy with overlap or diffuse energy, define a boundary by $\chi(T_c) = \chi_*$.

Proposition 7.1 (No universal wall-tap boundary; WT08 and WT09). *The certified delay separation depends on bandwidth, pulse shape, SNR, estimator and error criterion. The rhythm transition depends on the measured response tail and the declared statistic. For the exponential-tail control $\chi(T) = 1 - e^{-T/\tau_r}$ and $\chi_* = 1/2$,*

$$T_c = \tau_r \log 2.$$

Changing τ_r from 25 to 140 ms changes T_c by a factor 5.6. Consequently neither 30 ms nor any other single value is an RG constant.

What can lock is a typed certificate

$$\text{Cert}_{\text{overlap},W}(F) = \{T_c, \text{window}, \text{statistic}, \text{uncertainty}, \text{held-out result}\}.$$

This matches the typed-certificate discipline developed for chaotic and dirty experiments [6].

8 Uncertainty floor and held-out validation

Proposition 8.1 (Repetition floor; WT07). *If timing error is $\varepsilon_i = b + u_i$ with shared b and independent u_i , then*

$$\text{Var}(\bar{\varepsilon}_n) = \sigma_b^2 + \frac{\sigma_u^2}{n}.$$

Repetition cannot remove the medium, clock, device or model floor. The battery approaches the registered 35 μ s shared floor as n grows.

Proposition 8.2 (Held-out transport lock; WT10). *A calibration fitted on one range window may be transported only when it predicts held-out anchors. The registered affine device model gives sub-0.1% median held-out error; an injected nonlinear timing distortion increases the error by more than a factor three. Passing the fitted anchors alone is not certification.*

9 The Wall–Tap Physical Lock Theorem

Theorem 9.1 (Wall–Tap Physical Lock). *Let S_{WT} be a wall–tap physical slice and let W be a candidate data window. Suppose:*

1. *the trigger and clock channels define an operational duration object on W ;*
2. *source, detector and transducer latencies are identified or retained as nuisance fibres;*
3. *the acoustic path model is declared, and monostatic versus bistatic geometry is not conflated;*
4. *the medium slowness field is identified to the stated tolerance, or the procedure abstains from metres;*
5. *path assignments are locked by labels and geometric closure;*
6. *target and nuisance ranks are constant on W ;*
7. *device and response stationarity are certified by equivalence tests;*
8. *the resulting distance map predicts held-out anchors within the preregistered tolerance.*

Then the wall–tap record defines on W :

1. *a kernel-basic operational path duration;*
2. *a medium-anchored acoustic path length in metres;*
3. *on geometrically complete labelled branches, a local physical-distance chart;*
4. *a typed uncertainty and abstention certificate.*

If any of L1–L8 fails, the lower valid layer remains: event order, raw latency, calibrated duration, or integrated slowness. Failure never licenses promotion to physical distance.

Corollary 9.2 (Place on the complete ladder). *The wall–tap route is not a new primitive rung. It is a physical assembly path:*

- R1–R6 : joint acoustic law, kernel, quotient and saturation,*
- R28–R29 : clock kernel, cocycle and operational duration,*
- R38–R43 : scale fibre, medium anchor, integrability and held-out distance,*
- R45–R57 : joint-law naturality, stabilizer and cross-module locks.*

The acoustic branch refines the physical-distance hypotheses by naming the medium lock and the monostatic/bistatic path type.

10 Attack ledger

The executable battery `rg_wall_tap_lock_attack.py` uses seed 20260929; seeds 1, 42, 777 and 2026 repeat all checks.

ID	Registered claim	Verdict
WT01	raw latency versus calibrated path duration	pass
WT02	exact distance–speed fibre	no-go confirmed
WT03	medium anchor converts two-way duration to metres	pass
WT04	four labelled returns reconstruct a regular 2D chart	pass
WT05	unlabelled returns leave a permutation fibre	no-go confirmed

ID	Registered claim	Verdict
WT06	environmental speed error is common scale bias	pass
WT07	repetition retains a systematic floor	pass
WT08	delay boundary scales with pulse width and SNR	pass
WT09	rhythm transition scales with response tail	pass
WT10	held-out anchors detect nonlinear drift	pass
WT11	half-factor fails as wall distance in bistatic geometry	no-go confirmed
WT12	a hidden response change breaks the joint certificate	pass

The primary run passes 12/12. All four robustness seeds pass 12/12. The battery is synthetic and validates logical separation and numerical implementation only.

11 Non-claims

Non-claim 11.1. No historical timing precision is assigned to the childhood observation. Its scientific role is to specify the ordered-event, recurrence and rhythm hypotheses.

Non-claim 11.2. The paper does not reconstruct a room from one echo. One return supplies one path constraint; completeness requires independent paths, labels, scale and closure.

Non-claim 11.3. The acoustic speed is not the SI vacuum speed and is not derived by RG. It is an imported medium calibration with uncertainty and model scope.

Non-claim 11.4. The observation cone of the acoustic protocol is not a Lorentzian causal cone. No field equation, universal causal speed or new law of acoustics is derived.

Non-claim 11.5. The synthetic thresholds are registered battery choices, not performance guarantees for phones, rooms or human perception.

12 Conclusion

The mature physical wall–tap observation is neither “a child measured sound speed” nor “delay contains geometry.” Its stable content is more precise. A trigger precedes a return. Calibration turns raw latency into an operational duration. A path model turns duration into integrated slowness. A medium anchor turns slowness into metres. Labels and closure turn several path lengths into a geometry. Regularity, stationarity and held-out anchors determine where that geometry is certified. Rhythm supplies a history-law boundary whose value belongs to the response and protocol, not to RG as a universal constant.

The wall tap therefore locks the complete theory at exactly the intended place: it begins at the kernel of a joint observation law, climbs through operational duration, crosses the physical-distance scale only through an explicit medium anchor, and terminates in a windowed certificate with honest abstention. This is the mature physical version of the observation that originally motivated Resolution Geometry.

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